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🏠 Montpellier, France

📄 Driving License

☎ +33786614034

Languages

French (Native)

English (TOEIC: 880)

Interests

Acoustic and Electric Guitar
Sports

Online Profiles

🌐 [LinkedIn](#)

🔗 [Google Scholar](#)

Guillaume Fourret

PhD Computer Science – R&D Engineer

PhD in computer vision and deep learning, specialized in object detection and real-time drone video analysis. Strong interest in AI applications across various domains, including multimodal models and large language models.



Education

PhD in Computer Science

July 2026, University of Montpellier

Detection and in-depth analysis of regions of interest using AI algorithms in drone-acquired videos.

CIFRE (French industry-academia partnership) PhD in collaboration with the startup Drone Geofencing, LIRMM laboratory, ICAR team

Keywords: Deep Learning, Object Detection, Few-Shot Learning, Anomaly Detection, Real-Time Processing, Aerial Drones

Exchange Semester, Computer Science

August 2021 – December 2021, Polytechnique Montréal

Engineering Degree in Computer Science, Networks & Cybersecurity

2019 – 2022, Grenoble INP – ESISAR

With Honors

Preparatory Classes for Engineering Schools

2017 – 2019, Grenoble INP – ESISAR

Professional Experience

Master's Thesis Internship

February 2022 – July 2022, Drone Geofencing, Nîmes, France

Real-time detection, localization, and identification of objects of interest using Deep Learning on aerial drone imagery for long-endurance missions.

Industrial Project

January 2021 – July 2021, MGI Digital Technology, Lyon, France

Development of an interface driver and its testing GUI for an integrated color image scanner system.

Project in collaboration with a company within Grenoble INP – ESISAR.

Keywords: C++, CUDA, High-Speed Ethernet, TCP/UDP

Skills

● **Programming Languages:** Python, C++, CUDA (basic knowledge)

● **Frameworks & Libraries:** Pytorch, NumPy, OpenCV, JAX (basic knowledge)

Projects

Multimodal zero-shot object detection based on RF-DETR

Adapted RF-DETR into a multimodal detector that recognizes novel objects directly from their 3D CAD models, with no target-domain images required. Using a late-fusion design, the model learns a shared 3D-image latent space while preserving real-time inference. Pretrained on the large-scale synthetic MegaPose dataset via multi-GPU HPC (*Jean-Zay*) and evaluated on the real-world YCBV benchmark (*publication in progress*).

PinkCC Challenge

Second place in the PinkCC 2025 medical challenge on automatic segmentation of ovarian cancer tumors in CT medical images using AI methods.